Respiratory support strategy in adults with acute hypoxemic respiratory failure: A systematic review and network meta-analysis

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Supplementary e-Table 1 Search strategy a) PubMed search strategy (Performed on May 30, 2021)

#1	"Hypoxia"[mh] OR hypox*[tiab] OR "Respiratory Insufficiency"[mh] OR respiratory depression*[tiab] OR respiratory failure*[tiab] OR ventilatory depression*[tiab] OR respiratory insufficienc*[tiab] OR "Dyspnea"[mh] OR dyspnea*[tiab] OR "shortness of breath"[tiab] OR "Respiratory Distress Syndrome, Adult"[mh] OR acute respiratory distress[tiab] OR adult respiratory distress[tiab] OR respiratory distress syndrome*[tiab] OR RDS[tiab] OR ARDS[tiab] OR "Acute Lung Injury"[mh] OR acute lung injur*[tiab] OR ALI[tiab]
#2	"Respiratory Distress Syndrome, Newborn"[mh]
#3	#1 or #2
#4	"Noninvasive Ventilation"[mh] OR noninvasive ventilation*[tiab] OR non invasive ventilation*[tiab] OR NIV[tiab] OR NPPV[tiab] OR NIPPV[tiab] OR noninvasive positive pressure ventilation*[tiab] OR noninvasive mechanical ventilation*[tiab] OR noninvasive pressure support ventilation*[tiab] OR "Continuous Positive Airway Pressure"[mh] OR continuous positive airway pressure*[tiab] OR bilevel positive airway pressure*[tiab] OR biphasic positive airway pressure*[tiab] OR BIPAP[tiab]
#5	"Oxygen Inhalation Therapy"[mh] OR HFNC[tiab] OR HHFNC[tiab] OR HHHFNC[tiab] OR HFNO[tiab] OR HFNT[tiab] OR HFNOT[tiab] OR HFO[tiab] OR HFOT[tiab] OR NHF[tiab] OR NHFC[tiab] OR NHFT[tiab] OR NHFO[tiab] OR NHFOT[tiab] OR high flow therap*[tiab] OR high flow oxygen[tiab] OR nasal high flow[tiab]
#6	#4 or #5
#7	("Randomized Controlled Trial"[pt] OR "Controlled Clinical Trial"[pt] OR "Clinical Trials as Topic"[mh] OR randomized[tiab] OR placebo[tiab] OR randomly[tiab] OR trial[tiab] OR groups[tiab]) NOT (Animals [mh] NOT Humans [mh])
#8	#3 and #6 and #7

b) CENTRAL search strategy (Performed on May 30, 2021)

ID	Search
#1	[mh Hypoxia] OR hypox*:ti,ab OR [mh "Respiratory Insufficiency"] OR "respiratory depression":ti,ab OR "respiratory failure":ti,ab OR "ventilatory depression":ti,ab OR "respiratory insufficiency":ti,ab OR [mh Dyspnea] OR dyspnea:ti,ab OR "shortness of breath":ti,ab OR [mh "Respiratory Distress Syndrome, Adult"] OR "acute respiratory distress":ti,ab OR "adult respiratory distress":ti,ab OR "respiratory distress syndrome":ti,ab OR RDS:ti,ab OR ARDS:ti,ab OR [mh "Acute Lung Injury"] OR "acute lung injury":ti,ab OR ALI:ti,ab
#2	[mh "Respiratory Distress Syndrome, Newborn"]
#3	#1 OR #2
#4	[mh "Noninvasive Ventilation"] OR "noninvasive ventilation":ti,ab OR "non invasive ventilation":ti,ab OR NIV:ti,ab OR NPPV:ti,ab OR NIPPV:ti,ab OR "noninvasive positive pressure ventilation":ti,ab OR "noninvasive mechanical ventilation":ti,ab OR "noninvasive pressure support ventilation":ti,ab OR [mh "Continuous Positive Airway Pressure"] OR "continuous positive airway pressure":ti,ab OR "bilevel positive airway pressure":ti,ab OR "biphasic positive airway pressure":ti,ab OR BIPAP:ti,ab
#5	[mh "Oxygen Inhalation Therapy"] OR HFNC:ti,ab OR HHFNC:ti,ab OR HHHFNC:ti,ab OR HFNO:ti,ab OR HFNT:ti,ab OR HFNOT:ti,ab OR HFO:ti,ab OR HFOT:ti,ab OR NHF:ti,ab OR NHFC:ti,ab OR NHFT:ti,ab OR NHFO:ti,ab OR NHFOT:ti,ab OR "high flow therapy":ti,ab OR "high flow oxygen":ti,ab OR "nasal high flow":ti,ab
#6	#4 OR #5
#7	#3 AND #6
#8	[mh animals] NOT [mh humans]
#9	#7 NOT #8

c) Embase search strategy (Performed on May 30, 2021)

Set#	Searched for
S1	(EMB.EXACT("hypoxia") OR (EMB.EXACT("paroxysmal dyspnea") OR EMB.EXACT("dyspnea")) OR (EMB.EXACT.EXPLODE("respiratory failure")))
S2	(EMB.EXACT.EXPLODE("neonatal respiratory distress syndrome") OR EMB.EXACT.EXPLODE("acute lung injury") OR EMB.EXACT.EXPLODE("adult respiratory distress syndrome"))
S3	(TI,AB(hypoxi* OR (respiratory p/0 (depression* OR failure* OR insufficienc*)) OR (ventilatory p/0 depression*) OR dyspnea* OR "shortness of breath" OR "acute respiratory distress" OR "adult respiratory distress" OR "respiratory distress syndrome" OR RDS OR ARDS OR (acute p/0 lung p/0 injur*) OR ALI))
S4	(S1 or S2 or S3)
S5	(EMB.EXACT("noninvasive ventilation"))
S6	(TI,AB("noninvasive ventilation" OR "non invasive ventilation" OR NIV OR NPPV OR NIPPV OR "noninvasive positive pressure ventilation" OR "noninvasive mechanical ventilation" OR "noninvasive pressure support ventilation" OR "continuous positive airway pressure" OR "bilevel positive airway pressure" OR "biphasic positive airway pressure" OR BIPAP))
S7	(EMB.EXACT("home oxygen therapy") OR EMB.EXACT("hyperbaric oxygen therapy") OR EMB.EXACT("oxygen therapy"))
S8	(TI,AB(HFNC OR HHFNC OR HHHFNC OR HFNO OR HFNT OR HFNOT OR HFO OR HFOT OR NHF OR NHFC OR NHFT OR NHFO OR NHFOT OR ("high flow" p/0 (therap* OR oxygen)) OR "nasal high flow"))
S9	(S5 or S6 or S7 or S8)
S10	((EMB.EXACT("controlled clinical trial") OR EMB.EXACT.EXPLODE("clinical trial (topic)") OR EMB.EXACT("randomized controlled trial")) OR (TI,AB(randomized) OR TI,AB(randomly) OR TI(trial) OR TI,AB(groups)) NOT (ANIMAL(YES) NOT HUMAN(YES)))
S11	(S4 and S9 and S10)
S12	(S11 AND PD(>=2010))
S13	(S11 AND PD(<2010))

d) Ichushi search strategy (Performed on May 30, 2021)

Search	Query
#1	酸素欠乏/TH or 酸素欠乏/TA or anoxia/TA or Hypoxia/TA
#2	呼吸窮迫症候群-急性/TH or 急性呼吸窮迫症候群/TA or ARDS/TA
#3	急性肺損傷/TH or 急性肺損傷/TA or 急性肺障害/TA or 急性肺傷害/TA or ALI/TA
#4	呼吸不全/TH or 呼吸不全/TA
#5	呼吸困難/TH or 呼吸困難/TA
#6	呼吸窮迫症候群-新生児/TH or 新生児呼吸窮迫症候群/TA
#7	#1 or #2 or #3 or #4 or #5 or #6
#8	非侵襲的補助換気/TH or 非侵襲的補助換気/TA or NPPV/TA or NIPPV/TA
#9	持続気道陽圧/TH or 持続気道陽圧/TA or CPAP/TA
#10	非侵襲的陽圧呼吸/TH or 非侵襲的陽圧呼吸/TA or BIPAP/TA
#11	酸素吸入療法/TH or 酸素吸入/TA
#12	酸素療法/TA or ハイフロー/TA or HFNC/TA or NHF/TA or HFO/TA
#13	#8 or #9 or #10 or #11 or #12
#14	#7 and #13
#15	(#14) and (PT=会議録除く)
#16	ランダム化比較試験/TH or ランダム化/AL or 無作為化/AL
#17	比較試験/AL
#18	臨床試験/TH or 臨床試験/AL
#19	プラセボ/TH or プラセボ/AL
#20	対照/AL
#21	コントロール/AL
#22	臨床研究・疫学研究/TH or 臨床研究/AL
#23	#16 or #17 or #18 or #19 or #20 or #21 or #22
#24	#15 and #23

Supplementary e-Table 2 Risk of bias summary

(a) Short-term mortality

Source	Bias arising from the randomization process	Bias due to deviations from intended interventions	Bias due to missing outcome data	Bias in measurement of the outcome	Bias in selection of the reported result	Overall risk of bias
Wysock 1995	Low	Some concerns	Low	Low	Low	Some concerns
Antolnelli 1998	Some concerns	Some concerns	Low	Low	Low	High
Confalonieri 1999	Low	Some concerns	Low	Low	Low	Some concerns
Antonelli 2000	Low	Some concerns	Low	Low	Low	Some concerns
Delcaux 2000	Low	Some concerns	Some concerns	Low	Low	High
Martin 2000	Some concerns	Some concerns	Low	Low	Low	Hugh
Hilbert 2001	Low	Some concerns	Low	Low	Low	Some concerns
Ferrer 2003	Low	Some concerns	Low	Low	Low	Some concerns
Cosentini 2010	Low	Some concerns	Low	Low	Low	Some concerns
Squadrone 2010	Low	Some concerns	Low	Low	Low	Some concerns
Wermke 2012	Low	Some concerns	Some concerns	Low	Low	High
Zhan 2012	Low	Some concerns	Low	Low	Low	Some concerns
Brambilila 2014	Low	Some concerns	Low	Low	Low	Some concerns
Frat 2015	Low	Some concerns	Low	Low	Low	Some concerns
Lemiale2015_JAMA	Low	Some concerns	Low	Low	Low	Some concerns
Jones 2016	Low	Some concerns	Low	Low	Low	Some concerns
Muncharaz 2017	Low	Some concerns	Low	Low	Low	Some concerns
Azoulay 2018	Low	Some concerns	Low	Low	Low	Some concerns
He 2019	Low	Some concerns	Low	Low	Low	Some concerns
Andino 2020	Low	Some concerns	Low	Low	Low	Some concerns

AlptekİnoĞlu Mendİl 2021	Some concerns	Some concerns	Low	Low	Low	High
Awadallah 2021	Some concerns	Some concerns	Low	Low	Low	Some concerns
Groeco 2021	Some concerns	Some concerns	Low	Low	Low	High

(b) Endotracheal intubation

Source	Bias arising from the randomization process	Bias due to deviations from intended interventions	Bias due to missing outcome data	Bias in measurement of the outcome	Bias in selection of the reported result	Overall risk of bias
Wysock 1995	Low	Some concerns	Low	Low	Low	Some concerns
Confalonieri 1999	Low	Some concerns	Low	Low	Low	Some concerns
Antonelli 2000	Low	Some concerns	Low	Low	Low	Some concerns
Delcaux 2000	Low	Some concerns	Some concerns	Low	Low	High
Martin 2000	Some concerns	Some concerns	Low	Low	Low	High
Hilbert 2001	Low	Some concerns	Low	Low	Low	Some concerns
Ferrer 2003	Low	Some concerns	Low	Low	Low	Some concerns
Cosentini 2010	Low	Some concerns	Low	Low	Low	Some concerns
Squadrone 2010	Low	Some concerns	Low	Low	Low	Some concerns
Wermke 2012	Low	Some concerns	Some concerns	Low	Low	High
Zhan 2012	Low	Some concerns	Low	Low	Low	Some concerns
Brambilila 2014	Low	Some concerns	Low	Low	Low	Some concerns
Azevedo 2015	Some concerns	Some concerns	Low	Low	Low	High
Frat 2015	Low	Some concerns	Low	Low	Low	Some concerns
Lamiale 2015_CC	Low	Some concerns	Low	Low	Low	Some concerns
Lemiale2015_JAMA	Low	Some concerns	Low	Low	Low	Some concerns
Jones 2016	Low	Some concerns	Low	Low	Low	Some concerns
Azoulay 2018	Low	Some concerns	Low	Low	Low	Some concerns
He 2019	Low	Some concerns	Low	Low	Low	Some concerns
Andino 2020	Low	Some concerns	Low	Low	Low	Some concerns
AlptekİnoĞlu Mendİl 2021	Low	Some concerns	Low	Low	Low	Some concerns
Groeco 2021	Some concerns	Some concerns	Low	Low	Low	High

Supplementary e-Table 3 Sensitivity analysis excluding studies with P/F ratio ≥ 200

(a) Short-term mortality

Comparison	All studies-Main analysis			Excluding studies with P/F ratio ≥ 200		
	No. of studies	RR (95% CI)	Rating	No. of studies	RR (95% CI)	Rating
HFNO vs SOT	5	0.89 (0.66–1.20)	Low	5	0.65 (0.35–1.20)	Low
NIV vs SOT	15	0.76 (0.61–0.95)	Low	8	0.67 (0.42–1.05)	Low
IMV vs SOT	0	1.01 (0.57–1.78)	Very low	0	1.07 (0.41–2.79)	Low
HFNO vs IMV	0	0.89 (0.78–1.65)	Very low	0	0.61 (0.21–1.77)	Low
NIV vs IMV	3	0.76 (0.45–1.27)	Very low	3	0.62 (0.27–1.45)	Low
NIV vs HFNO	2	0.85 (0.61–1.20)	Very low	2	1.03 (0.53–1.99)	Low

Testing for global incoherence: Overall RCTs, P= 0.01; Excluding studies with P/F ≥ 200, P=0.13.

(b) Endotracheal intubation

Comparison	All patients-Main analysis			Excluding studies with P/F ratio ≥ 200		
	No. of studies	RR (95% CI)	Rating	No. of studies	RR (95% CI)	Rating
HFNO vs SOT	6	0.89 (0.61–1.11)	Moderate	4	0.77 (0.54–1.10)	Low
NIV vs SOT	15	0.63 (0.51–0.79)	Low	8	0.62 (0.48–0.81)	Low
NIV vs HFNO	3	0.77 (0.56–1.06)	Moderate	2	0.80 (0.55–1.17)	Low

Testing for global incoherence: Overall RCTs, P= 0.09; Excluding studies with P/F ≥ 200, P=0.09

Supplementary e-Table 4 Sensitivity analysis excluding studies with Type II respiratory failure

(a) Short-term mortality

Comparison	All studies-Main analysis			Excluding studies with Type II respiratory failure		
	No. of studies	RR (95% CI)	Rating	No. of studies	RR (95% CI)	Rating
HFNO vs SOT	5	0.89 (0.66–1.20)	Low	3	0.73 (0.43–1.25)	Low
NIV vs SOT	15	0.76 (0.61–0.95)	Low	11	0.69 (0.49–0.98)	Low
IMV vs SOT	0	1.01 (0.57–1.78)	Very low	0	0.94 (0.45–1.93)	Very low
HFNO vs IMV	0	0.89 (0.78–1.65)	Very low	0	0.78 (0.45–1.82)	Very low
NIV vs IMV	3	0.76 (0.45–1.27)	Very low	3	0.75 (0.39–1.40)	Very low
NIV vs HFNO	2	0.85 (0.61–1.20)	Very low	2	0.94 (0.54–1.64)	Very low

Testing for global incoherence: Overall RCTs, P= 0.01; Excluding studies with Type II respiratory failure, P=0.17.

(b) Endotracheal intubation

Comparison	All patients-Main analysis			Excluding studies with Type II respiratory failure		
	No. of studies	RR (95% CI)	Rating	No. of studies	RR (95% CI)	Rating
HFNO vs SOT	6	0.89 (0.61–1.11)	Moderate	4	0.81 (0.50–1.32)	Low
NIV vs SOT	15	0.63 (0.51–0.79)	Low	11	0.61 (0.44–0.85)	Low
NIV vs HFNO	3	0.77 (0.56–1.06)	Moderate	2	0.75 (0.46–1.24)	Low

Testing for global incoherence: Overall RCTs, P= 0.30; Excluding studies with Type II respiratory failure, P=0.42

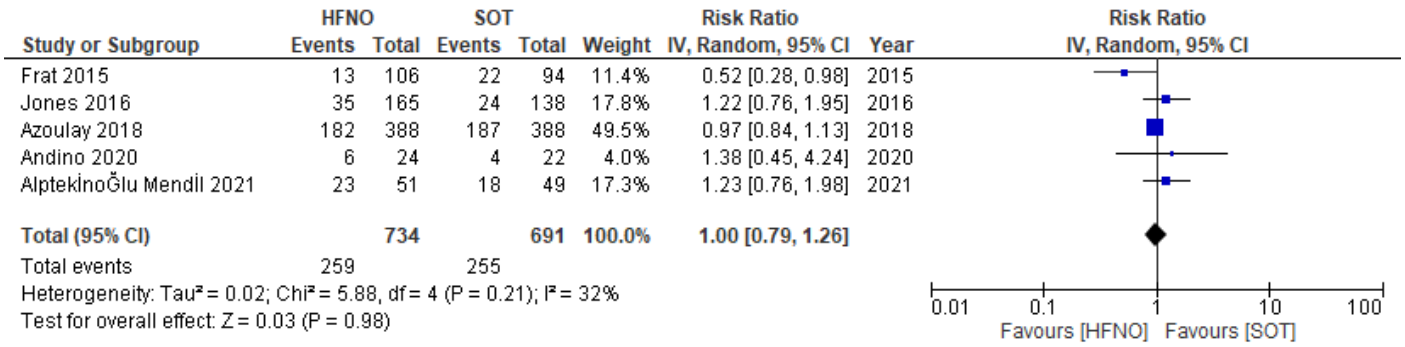
Supplementary e-Table 5 Sensitivity analysis for mortality excluding studies that reported long-term mortality

Comparison	All studies-Main analysis			Excluding studies with long term mortality		
	No. of studies	RR (95% CI)	Rating	No. of studies	RR (95% CI)	Rating
HFNO vs SOT	5	0.89 (0.66–1.20)	Low	2	1.26 (0.69–2.31)	Very low
NIV vs SOT	15	0.76 (0.61–0.95)	Low	12	0.64 (0.49–0.83)	Very low
IMV vs SOT	0	1.01 (0.57–1.78)	Very low	0	0.84 (0.47–1.47)	Very low
HFNO vs IMV	0	0.89 (0.78–1.65)	Very low	0	1.50 (0.65–3.48)	Very low
NIV vs IMV	3	0.76 (0.45–1.27)	Very low	3	0.76 (0.45–1.27)	Very low
NIV vs HFNO	2	0.85 (0.61–1.20)	Very low	0	0.50 (0.54–3.79)	Very low

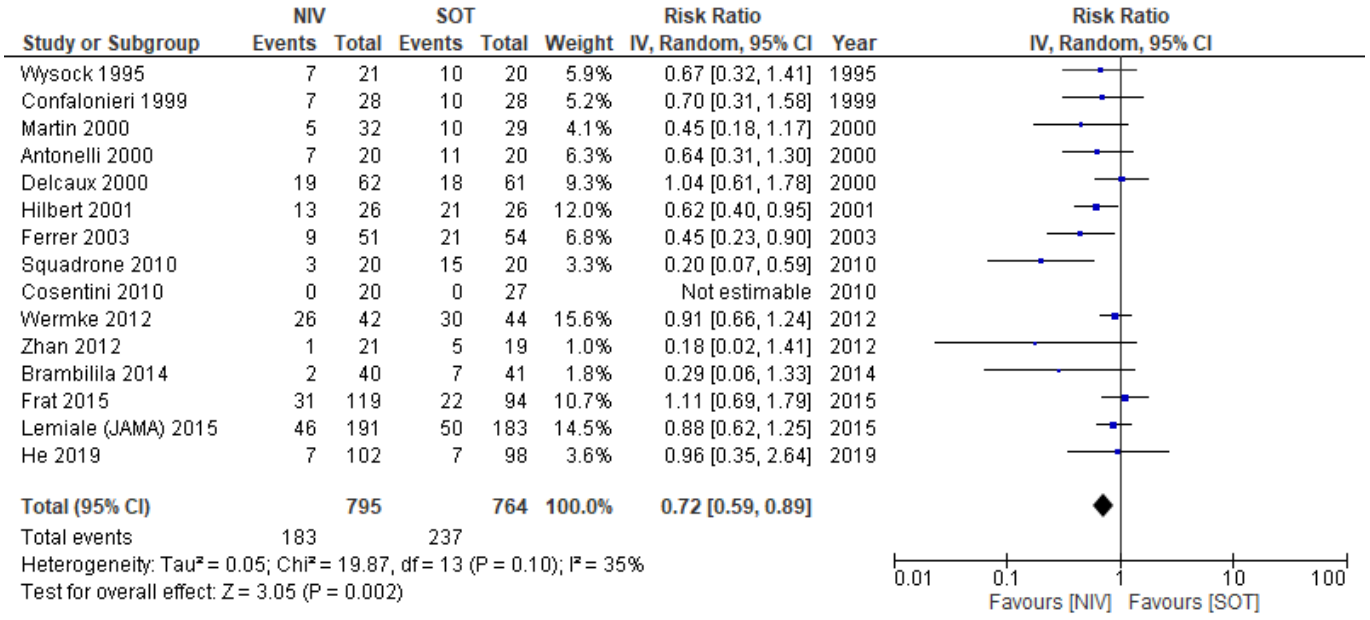
Testing for global incoherence: Overall RCTs, P= 0.01; Excluding studies with long mortality, P=0.06.

Supplementary e-Fig. 1 Forest plots for the pairwise comparison of short-term mortality

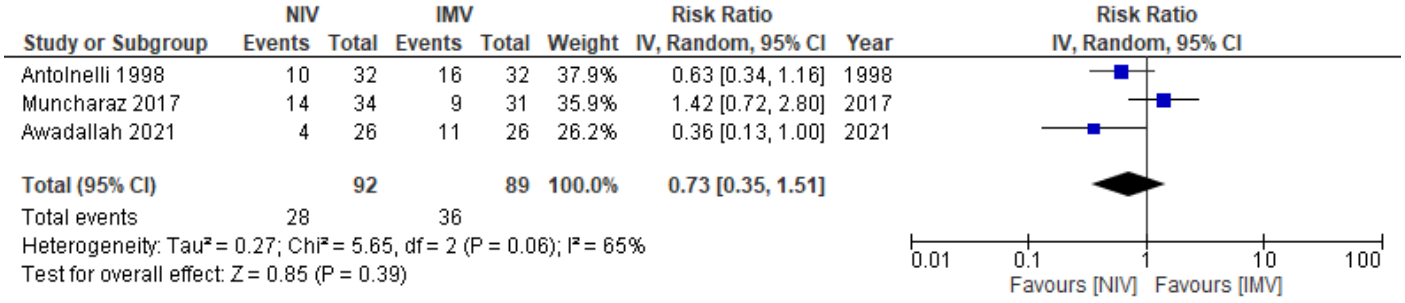
(a) HFNO vs SOT



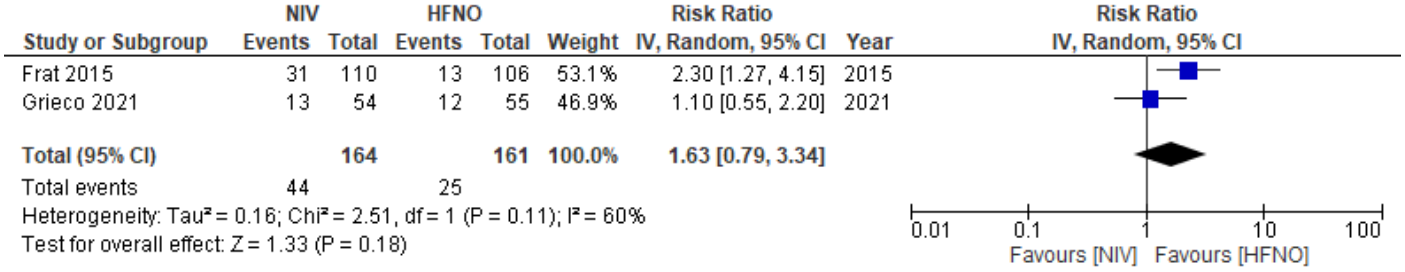
(b) NIV vs SOT



(c) NIV vs IMV

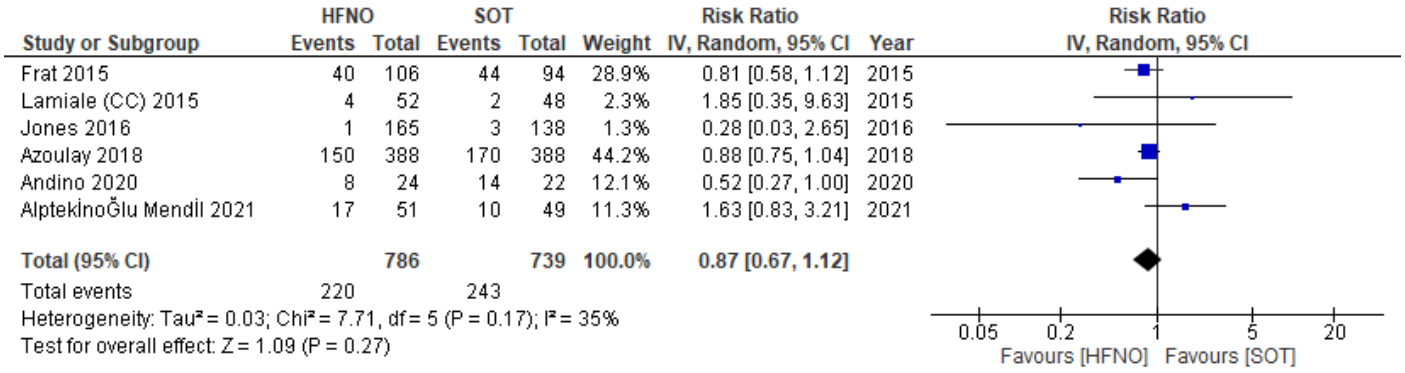


(d) NIV vs HFNO

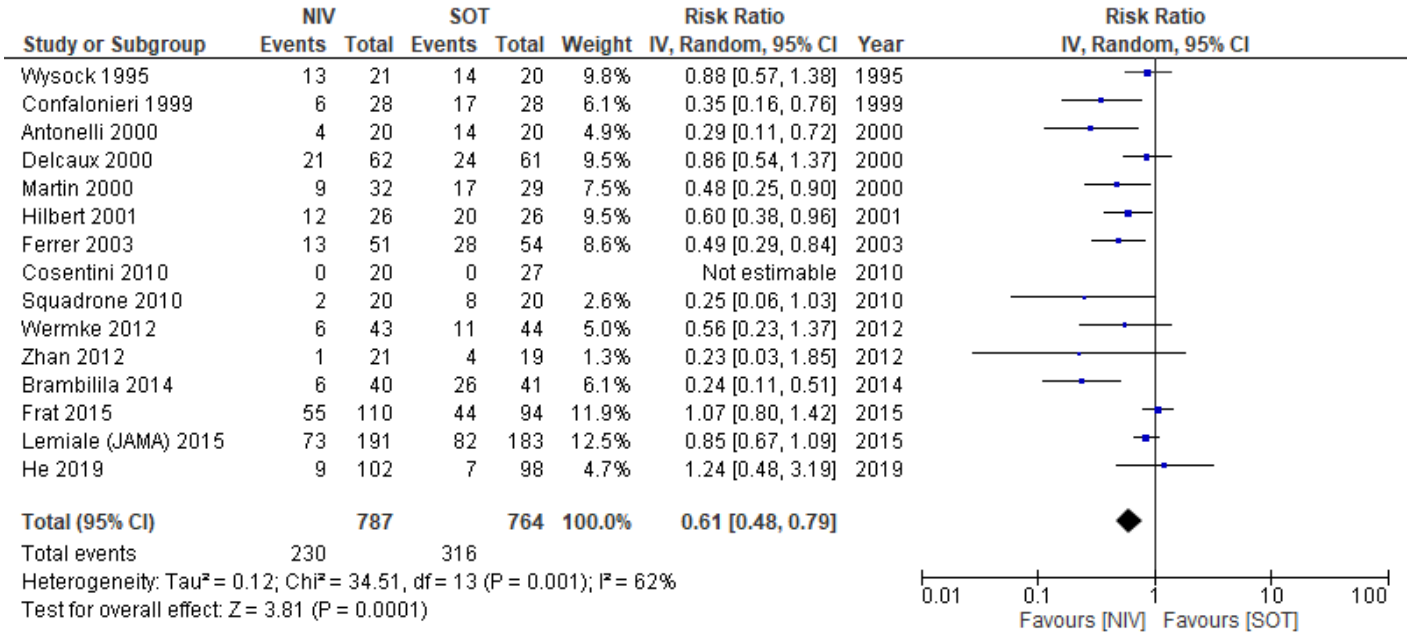


Supplementary e-Fig. 2 Forest plots for the pairwise comparison of endotracheal intubation

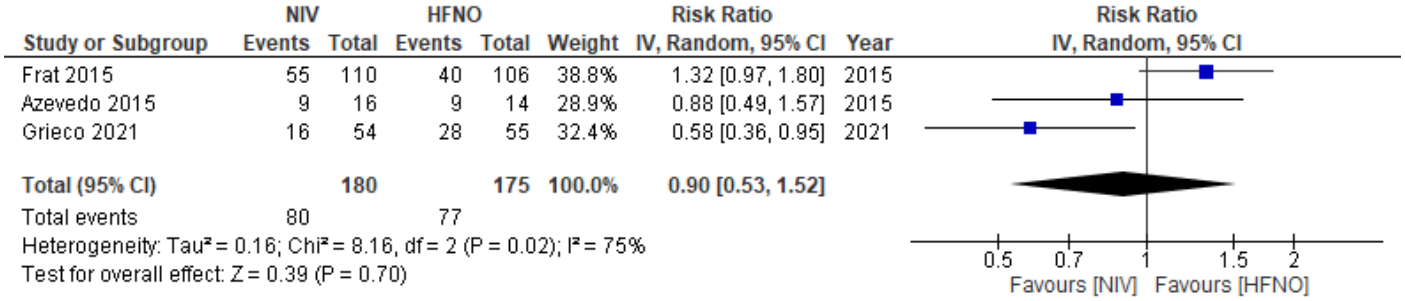
(a) HFNO vs SOT



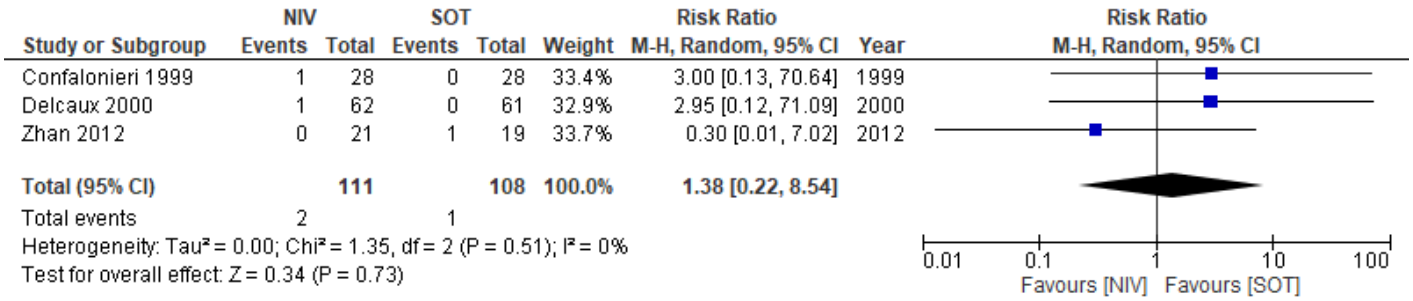
(b) NIV vs SOT



(c) NIV vs HFNO

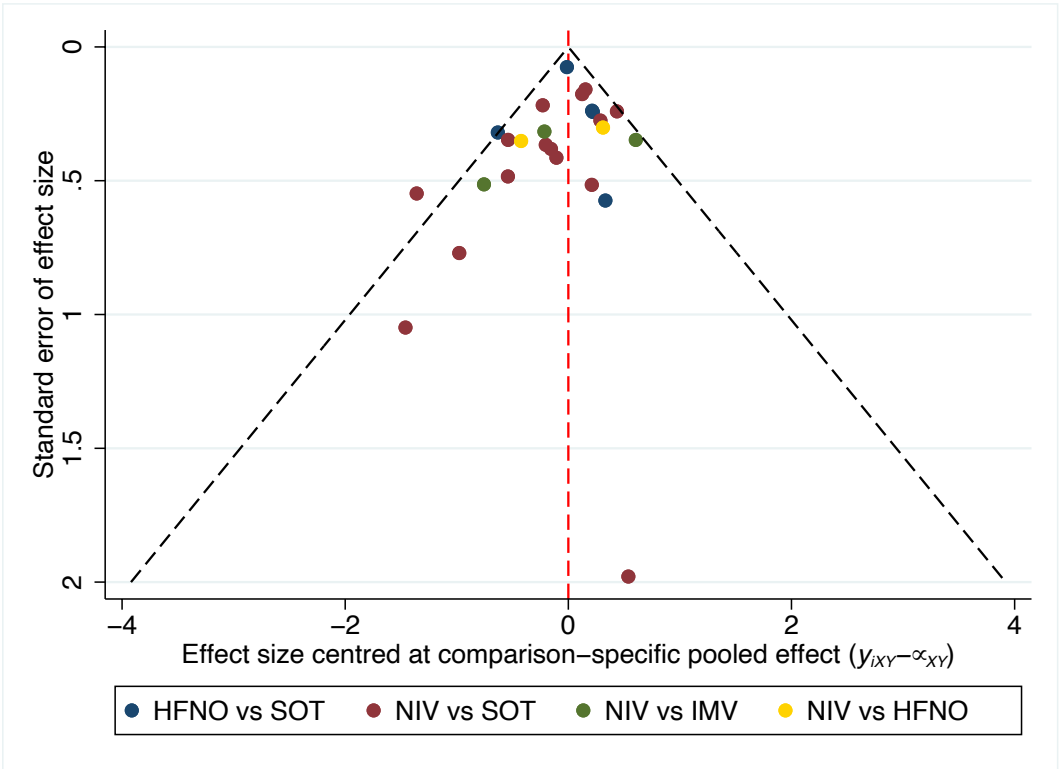
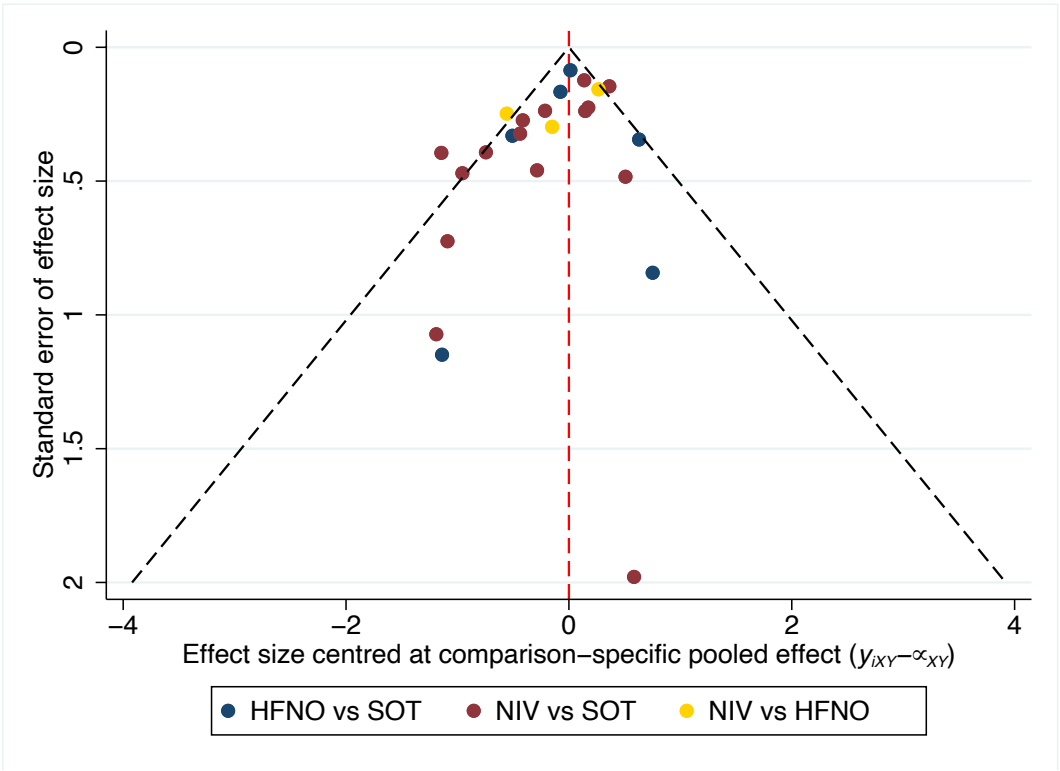


Supplementary e-Fig. 3 Forest plots for the pairwise comparison of ventilator associated lung injury



Supplementary e-Fig. 4 Funnel plots for each outcome

(a) Short-term mortality



(b) Endotracheal intubation